GAMIFICATION IN HUMAN RESOURCE MANAGEMENT: ENHANCING EMPLOYEE ENGAGEMENT AND MOTIVATION

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DEDICATION

We dedicate this project to our beloved parents, whose unwavering support has been the cornerstone of our educational journey. Their sacrifices, encouragement, and boundless love have provided us with the strength and motivation to pursue our dreams.

Every late night spent studying, every challenge faced, and every milestone achieved has been made possible by their steadfast belief in our abilities. They have instilled in us the values of perseverance and hard work, guiding us through both triumphs and obstacles.

As we prepare to graduate, we honor them for their relentless dedication and commitment, which have shaped us into the individuals we are today. This accomplishment is not just ours; it belongs to them as well. Thank you for being our steadfast pillars of support and for always believing in our potential.

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We are truly thankful for the time and effort all our mentors invested in us, as it has profoundly impacted our academic endeavor and personal growth. With their mentorship, we have not only achieved our objectives but have also cultivated a lasting passion for learning.

DECLARATION

We hereby certify that this material, which I now submit for assessment on the programme of study leading to the award of Bachelor of Science in *Bachelor of Science in Information Technology* is entirely my own work, that I have exercised reasonable care to ensure that the work is original, and does not to the best of my knowledge breach any law of copyright, and has not been taken from the work of others save and to the extent that such work has been cited and acknowledged within the text of my work.

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ABSTRACT

Employee disengagement remains a critical challenge in modern organizations, with approximately 70% of staff experiencing demotivation. Traditional Human Resource Management (HRM) tools, such as annual reviews and conventional training, often fail to sustain long-term engagement, particularly among the younger workforce (Generation Z and Millennials). To address this, this project investigates **Gamification** – the application of game-design elements like points, badges, and leaderboards in non-game contexts – as a strategic framework to enhance motivation and professional development. Using **Self-Determination Theory (SDT)** as a psychological foundation, the project explores how gamified HR systems can fulfil core needs for autonomy, competence, and relatedness. The study aims to identify the mechanisms through which gamification activates intrinsic motivation and evaluates its effectiveness across different employee demographics. The research progresses through a structured methodology, including a background study, literature review, system requirements analysis, and the development of a gamified implementation framework. Ultimately, this work demonstrates that gamification can transform the routine workplace into a highly engaging, productive environment in alignment with digital transformation goals.

Table of Contents

DEDICATION *2*

ACKNOWLEDGMENT *3*

DECLARATION *4*

ABSTRACT *5*

Chapter 1 ..... *10*

Introduction *10*

1.2 Problem statements **11**

1.3 Aim and Objectives **11**

- Research Aim 11
- Research Objectives 11

1.4 Significance of Research **12**

CHAPTER 2: *13*

THEORETICAL BACKGROUND & CONCEPTUAL FRAMEWORK *13*

2.1 Gamification and the Modern Workplace **13**

2.2 Theoretical Framework: Self-Determination Theory (SDT) **13**

2.3 Key Gamification Mechanics (The PBL Triad) **13**

2.4 Integration in HRM Pillars **14**

CHAPTER 3 ..... *15*

LITERATURE REVIEW *15*

3.1 Global Trends in Gamified HR Systems **15**

3.2 Impact on Motivation and Behavioral Change **15**

3.3 Challenges and Implementation Barriers **15**

CHAPTER 4 ..... *16*

SYSTEM DESIGN AND METHODOLOGY *16*

4.1 Functional Requirements **16**

4.2 Non-Functional Requirements **17**

4.3 Proposed System Architecture: Decentralized Intelligence **19**

4.4 Research Methodology **19**

4.4.1 Research Design **19**

4.4.2 Conceptual Framework **19**

4.4.3 Population and Sampling 19

4.4.4 Data Collection Instrument 19

4.4.5 Data Analysis Plan.....20

4.4.6 Ethical Considerations.....20

LIST OF FIGURES

Number of Figure	Figure Description

LIST OF ABBREVIATION

Acronym	**Definition of Acronym**
Gen z	Generation Z
SDT	Self-determination theory
HR	Human Resources
IU	Islamic University
HRM	Human Resources Management
PBL	Point Badge Leaderboard
FR	Functional Requirement
NFR	Non-Functional Requirement

Chapter 1

Introduction

Gamification involves the application of game-designed tools and mechanics in a non-game context, especially within an organization, to improve employee motivation and engagement [1]. These elements include point systems, leaderboards, badges, progress tracking, challenges, and reward mechanisms, which are introduced into human resources management processes such as recruitment, performance evaluation, and employee training and development. In the contemporary organizational context, employee disengagement and demotivation remain significant challenges, with about 70% of employees experiencing disengagement. Gamification has emerged as a strategic approach to transform the routine workplace into an engaging and motivated environment with rewarding experiences.[2, 3]

The significant importance of gamification in human resource management is affected by different factors. Traditional engagements such as annual performance review, financial incentives, and conventional recognition programs usually fail to sustain employee motivation and foster significant engagement.[4] However, the modern workplace requires novel approaches that address not just immediate metrics but also include long-term employee satisfaction, retention, and professional development. Investment in employee development remains underutilized. The organization spends a huge number of resources on training and development programs yet fails to meet the objective of behavioural change among unmotivated employees or skill sustenance application in workplace context. The rate of training completion and knowledge retention usually falls short of organizational expectations, showing that traditional delivery methods may not adequately address contemporary learning preferences and motivational dynamics[5].

This challenge is, however, prevalent for the younger workforce (Gen Z and millennial) who fundamentally demonstrate preferences in their engagement compared to previous generations, displaying a high affinity for interactive, technology-enabled learning experiences and game-like learning.[5, 6] While studies have significantly increased on the application of game-designed elements in non-game contexts, demonstrating significant promise for improving employee engagement motivation, a critical grey area still exists regarding how gamification typically influences the motivation of employees and the engagement mechanism. The underlying psychological process via which gamification-designed tools activate and sustain intrinsic motivation still needs deeper examination on how

effects operate primarily through fulfilment as self – determination theory predicts, or via alternative mechanisms such as achievement motivation, social comparison, or novel effects. Understanding the detailed mechanism would improve targeted and effective implementation.[7] Also, there is a limitation of insight into variation in effectiveness across employee populations, either by age, role, or individual characteristics, and job type, which is essential for targeted implementation strategies.

1.2 Problem statements

Organizations are confronted with enormous challenges in sustaining and activating employee motivation, engagement, and professional development while currently addressing workforce preferences for interactive and personalized experiences. Gamification is a promising approach with documented potential to address these challenge nexuses, offering game-design principles to improve intrinsic motivation, foster workplace interaction, and enhance skills development. However, there is insufficient empirical understanding regarding the psychological mechanisms through which gamification influences employee motivation and engagement; design principles that examine implementation success across wide organizational context; organizational barrier that impede effective implementation.

This gap creates a critical opportunity for further research on investigating gamification mechanisms, moderating factors, and implementation success factors that will enhance the advancement of organizations to understand how to effectively deploy this emerging practice while contributing to wider human resource management and organizational behaviour theory.

1.3 Aim and Objectives

- **Research Aim**

The primary aim of this project is to evaluate the effectiveness of gamification in enhancing employee motivation and engagement at the Islamic University of Madinah, with a focus on identifying the psychological drivers and organizational barriers unique to this institutional environment.

- **Research Objectives**

To achieve this aim, the study will focus on the following specific objectives:

1. **To identify** the most effective gamification mechanics (points, badges, leaderboards, etc.) currently preferred by the university employees.

2. **To analyse** the impact of gamified elements on intrinsic motivation using the Self-Determination Theory framework (Autonomy, Competence, and Relatedness)
3. **To compare** the engagement levels between different demographic groups (e.g., Gen Z/Millennials vs. older employees) within the university.
4. **To propose** a set of implementation strategies to overcome organizational barriers identified during the research.

1.4 Significance of Research

This investigation will contribute to human resource management in IU, organizational behaviour, and motivation theory by investigating how game-designed principles will activate primary psychological needs and intrinsic motivation in the Islamic University employees. This finding will advance self-determination theory application in workplace environment and clarify mechanism through which organizational norms affect employee motivation.

CHAPTER 2:

THEORETICAL BACKGROUND & CONCEPTUAL FRAMEWORK

2.1 Gamification and the Modern Workplace

In the contemporary organizational context, traditional engagement methods are becoming obsolete. The modern workplace requires novel approaches to address long-term employee satisfaction and retention. Gamification has emerged as a strategy to transform routine tasks into rewarding experiences by introducing game-designed tools into Human Resource Management (HRM) processes, such as recruitment, onboarding, and performance evaluation.

2.2 Theoretical Framework: Self-Determination Theory (SDT)

The core of gamification's success lies in the fulfillment of psychological needs. Self-Determination Theory (SDT) predicts that when employees feel a sense of Autonomy (control), Competence (mastery), and Relatedness (connection), their intrinsic motivation is activated.

Gamified systems serve as the technological bridge to activate these psychological triggers, ensuring that employees are not just working for a paycheck, but for a sense of personal achievement.

2.3 Key Gamification Mechanics (The PBL Triad)

Effective gamification is built on the PBL Triad, which provides the structural “hooks” for engagement:

Points: Provide instant feedback on daily tasks and quantify progress.

Badges: Visual representations of skill attainment, certifications, and milestones.

Leaderboards: Utilize social comparison to foster healthy competition and visibility.

Game Mechanic	Psychological Need (SDT)	Organizational Impact
Quests / Challenges	Psychological Need (SDT)	Employees choose their "path," increasing ownership of work.
Points / Levels	Competence	Provides immediate proof of skill growth and mastery.
Leaderboards / Kudos	Relatedness	Builds community through social recognition and competition.

2.4 Integration in HRM Pillars

Gamification is no longer limited to simple training; it is integrated across the employee lifecycle:

- **Recruitment:** Using game-based assessments to screen candidates for cognitive fit.
- **Training (L&D):** Boosting completion rates through interactive progress tracking.
- **Performance Management:** Replacing static annual reviews with real-time “level-ups” and continuous feedback.

CHAPTER 3

LITERATURE REVIEW

3.1 Global Trends in Gamified HR Systems

Studies show that 70% of employees struggle with disengagement. Recent systematic reviews (Prianto et al., 2025) indicate a trend toward "Digital HR Systems" that use game elements to appeal to younger generations (Millennials and Gen Z) who have a high affinity for technology-enabled learning and instant gratification.

3.2 Impact on Motivation and Behavioral Change

Research (Zhang et al., 2026) demonstrates that gamified HRM acts as a primary driver of engagement through intrinsic motivation. Unlike financial incentives, which can sometimes decrease long-term interest, gamification fosters "achievement motivation," where the primary reward is the mastery of the skill itself.

3.3 Challenges and Implementation Barriers

Despite the promise, barriers exist. Gandrita et al. (2024) identify that if gamification is implemented without a human-centric design, it can lead to “fatigue” or “stress.” A critical gap remains in understanding how these tools sustain motivation over long periods beyond the initial “novelty effect.”

CHAPTER 4

SYSTEM DESIGN AND METHODOLOGY

4.1 Functional Requirements

These define the core operations the gamified system must perform. They are broken down by module to ensure clarity:

1) **Gamification Engine (PBL & Mechanics)**
    - **FR1: Points System:** The system shall automatically award Experience Points (XP) to employees upon completion of specific tasks (e.g., completing a training module, submitting a report on time, achieving 100% attendance for a week).
    - **FR2: Badge Management:** The system shall issue digital badges for specific milestones (e.g., "Master of Efficiency," "Team Player," "5-Year Service Anniversary") that are displayed on the user's profile.
    - **FR3: Dynamic Leaderboards**: The system shall generate department-specific and university-wide leaderboards. To maintain fairness and avoid discouragement (addressing "fatigue" mentioned in 3.3), leaderboards must have filters (e.g., "This Week," "This Month," "Best Progress").
    - **FR4: Quest/Challenge System**: The system shall allow HR managers to create "Daily Quests" (quick tasks) and "Epic Challenges" (long-term projects) that employees can voluntarily accept to satisfy the need for Autonomy.
2) **User managemant and Progress Managemant**
    - **FR5: Comprehensive User Profiles:** The system shall maintain a profile for each employee displaying:
        - Current Level/Rank.
        - Total XP accumulated.
        - Badge showcase.
        - Progress bar to the next level.
    - **FR6: Skill Tree / Competence Map:** To satisfy the need for Competence, the system shall visualize a "Skill Tree" showing the employee's current competencies and the requirements (XP/Badges needed) to master the next skill.

- **FR7: Feedback & Notification System:** The system shall provide real-time feedback. When a task is completed, the user must receive an instant notification: "Congratulations! You earned +50 XP."

3) **Social and collaborative features**
    - **FR8: Peer-to-Peer Recognition (Kudos):** The system shall allow employees to send "Kudos" or virtual tokens of appreciation to colleagues for help. This directly targets the Relatedness need.
    - **FR9: Team Formation:** The system shall support the creation of squads or teams for departmental challenges, allowing team-based leaderboards to foster collaboration over pure competition.
4) **Administrative and Reporting**
    - **FR10: Demographics Filtering:** The system must include an admin panel allowing researchers/HR to filter engagement data by Age Group (Gen Z, Millennial, Gen X) and Job Role (Academic/Admin) to satisfy Objective 3 of the research.
    - **FR11: Survey Integration:** The system should be capable of triggering the "Basic Psychological Need Satisfaction at Work Scale" (mentioned in 4.8) at regular intervals to correlate gamification usage with SDT scores.

4.2 Non-Functional Requirements

These define the quality attribute of the system and how well it performes the functions:

1) **Usability and Accessibility**
    - **NFR1: Generational Accessibility (User-Friendly Interface):** Given the diverse demographic (Objective 3), the UI must be intuitive. While a Gen Z employee might navigate complex menus quickly, the interface must remain simple and readable for older generations (Gen X). Tooltips and tutorials should be available.
    - **NFR2: Mobile Responsiveness:** The system must be fully functional on mobile devices (smartphones and tablets) to allow staff to check leaderboards or receive notifications on the go.
    - **NFR3: Multilingual Support:** To suit the context of the Islamic University of Madinah (which hosts international staff/students), the interface should support both English and Arabic seamlessly.

2) **Performance and Reliability**
    - **NFR4: Real-Time Accuracy (Low Latency):** As stated in the "Decentralized Intelligence" model (4.3), the system must update points and leaderboards in real-time (< 1-second delay). If a user completes a task and the XP doesn't show up immediately, trust in the system breaks.
    - **NFR5: Concurrent User Support:** The system must maintain performance during peak usage (e.g., end-of-month reporting or when a popular challenge ends) without crashing. Target: Support 95% of staff online simultaneously.
3) **Securitiy and Privacy**
    - **NFR6: Data Anonymization for Research:** To comply with the Ethical Considerations (4.9) , the system must have a backend view for researchers where data is anonymized. While admins see "Jalaal Deen," the research export should see "User_411028693" to ensure privacy.
    - **NFR7: Role-Based Access Control (RBAC):** Strict permissions must be enforced.
        - **Employee:** Can see own profile and public leaderboards.
        - **Manager:** Can see team progress.
        - **HR/Researcher:** Can see analytics but not edit personal data.
        - **Admin:** Full technical control.
4) **Maintainability and Scalability**
    - **NFR8: Modular Architecture:** The system should be built modularly (e.g., separate modules for Points, Badges, Social). This allows the university to fix bugs or update the "Badge" logic without shutting down the entire "Points" system.
    - **NFR9: Audit Trail:** The system must log all point transactions. If an employee disputes their score, an admin can view the immutable log of why they got those points.
5) **Cultural and Regulatory Complience**
    - **NFR10: Cultural Alignment:** Since the research is specific to the Islamic University of Madinah, the design (themes, badges, communication style) must respect the local culture and values of the

institution to prevent the "cultural resistance" mentioned in the barriers (4.8).

4.3 Proposed System Architecture: Decentralized Intelligence

The system uses a Decentralized Intelligence Model to ensure real-time responsiveness. By processing data closer to the user interaction, the system avoids the "lag" found in traditional databases, ensuring that when a task is completed, the reward (XP/Points) is reflected instantly.

4.4 Research Methodology

4.4.1 Research Design

This study will employ a descriptive and correlational research design using a quantitative approach. This design is appropriate because the study seeks to measure the relationship between specific gamification mechanics (independent variables) and employee motivation/engagement (dependent variables).

4.4.2 Conceptual Framework

The study is anchored in the Self-Determination Theory (SDT). The methodology will test how gamified elements satisfy the three innate psychological needs:

- **Autonomy:** The feeling of choice and control over tasks.
- **Competence:** The sense of mastery and skill acquisition.
- **Relatedness:** The connection to others through social mechanics (leaderboards/teams).

4.4.3 Population and Sampling

- **Target Population:** All administrative and academic staff at the Islamic University.
- **Sampling Technique:** Stratified Random Sampling. This ensures representation across different generations (Gen Z, Millennials, Gen X) and roles (Academic vs. Administrative), which directly addresses Objective 3.
- **Sample Size:** (Insert number, e.g., 150–200 participants) determined using a margin of error (e.g., 5%).

4.4.4 Data Collection Instrument

A structured online questionnaire will be developed, divided into four sections:

- **Demographics:** Age, gender, job role, and years of experience.

- **Gamification Preferences:** A Likert scale (1–5) to rank preferences for points, badges, leaderboards, and challenges (Objective 1).
- **Psychological Impact (SDT Scale):** Utilizing the Basic Psychological Need Satisfaction at Work Scale to measure Autonomy, Competence, and Relatedness (Objective 2).
- **Barriers & Strategies:** Open-ended and multiple-choice questions regarding organizational hurdles (e.g., lack of tech, cultural resistance) (Objective 4).

4.4.5 Data Analysis Plan

To ensure the data is converted into meaningful insights, the following statistical tests will be used:

- **Descriptive Statistics:** (Mean, Frequency, Standard Deviation) to identify the most preferred mechanics.
- **T-Tests/ANOVA:** To compare engagement levels across different age groups and departments (Objective 3).

Correlation Analysis (Pearson's r): To determine the strength of the relationship between gamification elements and intrinsic motivation.

4.4.6 Ethical Considerations

As the study involves university employees, the following ethics will be maintained:

- **Informed Consent:** Participants will be briefed on the study's purpose.
- **Anonymity:** No personal identifiers will be collected to ensure honest feedback regarding organizational barriers.
- **Right to Withdraw:** Participation is strictly voluntary.

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